Achilleas Santi Seisa

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Summary

- Roboticist and networked control systems researcher specializing in edge/cloud-enabled autonomous systems, autonomous mobile robots, artificial intelligence, and communication-control-computation co-design. Research experience spans scalable and resilient robotic architectures, dynamic resource allocation, and industrial autonomous systems, with contributions across Swedish and EU-funded research projects, academic publications, and industrial collaborations.

Professional Experience

Present — **Experienced Researcher** at **Ericsson Research**, Cyber Physical Systems, Decision & Control, Stockholm, Sweden (since Dec. 2025)

- *Development of concepts for solving challenges in communication-control-computation co-design*
- *Evaluation and demonstration of solutions and algorithms in testbed environments*
- *Publishing research through patents, top-tier conferences and journals*
- *Supervision of M.Sc. students and Ph.D. interns*
- *Engagement in internal and external research collaborations for driving the future of connected devices*

2025 — **CTO | Co-Founder** at **Eighth Dimension GmbH**, Munich, Germany | *Robotics startup*

- *Module integration for detection and tracking of objects in real time*
- *Module integration for multi-modal and multi-session 3D SLAM*
- *Development of navigation and planning tools utilizing continuously updated 3D world models*

Visiting Researcher at **Epiroc**, Automation Lab, Örebro, Sweden

- *Facilitate research directions on dynamic resource allocation for edge-enabled autonomous machines*

2023 — **Visiting Researcher** at **Ericsson Research**, Cloud and Technology, Lund, Sweden

- *Development of Kubernetes-based scalable cloud-robot architectures for multi-agent systems*

2021 — **Automation Engineer** at **Rockwell Automation**, (I.C. SYSTEMS), Athens, Greece

- Tasty Foods S.A.: *Development of SCADA systems*
- V. Bougatsos S.A.: *Design and implementation of electrical panels | Development of SCADA systems*
- Motor Oil (Hellas), Corinth Refineries S.A.: *Development of PLC systems | Design of electrical panels*

Education

2025 — **Ph.D.** at Luleå University of Technology, Robotics and AI research group, Luleå, Sweden
Thesis title: *Towards the Next Generation Robotic Autonomy on the Edge: Scalability, Resiliency, and Dynamic Resource Allocation*

2023 — **Licentiate** at Luleå University of Technology, Robotics and AI research group, Luleå, Sweden
Thesis title: *Towards Enabling the Next Generation of Edge Controlled Robotic Systems*

2020 — **Integrated M.Sc.** at University of Patras, Electrical and Computer Engineering, Patras, Greece
Thesis title: *Development and Control of an Advanced Robotic Tool for Enhanced Minimally Invasive Robotic Surgery*

Skills

Languages	English | Greek
Coding	Python | MATLAB | Simulink | C | C++
Robotics	ROS | ROS2 | CAD
Networks	5G | MQTT | TCP/IP
Cloud/Edge	Docker | Kubernetes | Helm
Industry	PLC | SCADA | HMI | Electrical Design
Misc.	Academic Research | Teaching | Supervising | Project Management

Projects

2024
- **SUNRISE-6G,** EU's Horizon Research and Innovation Programme, Ph.D. Researcher (1 year)
  Project: *Edge-offloading and data processing for enhancing autonomous multi-agent robotic missions*
  Team coordinator for the Robotics and AI group
- **AMBITIOUS,** EU's Horizon Research and Innovation Programme, Ph.D. Researcher (1 year)
  Project: *Autonomous UAV missions for IoT sensor data collection and operation*
  Team coordinator for the Robotics and AI group
- **PERSEPHONE,** EU's Horizon Research and Innovation Programme, Ph.D. Researcher (1 year)
  Project: *Server-robot communication for enhanced autonomous robotic mining-drilling missions*
- **ARCTIC EDGE,** EU's Northern Periphery and Arctic Programme, Ph.D. Researcher (1 year)
  Project: *Implementation of a cloud-based, IoT, and MQTT technology-driven solution for rural manufacturing NPA SMEs, aiming to optimize supply chain processes*
- **LIFELONG LEARNING,** Luleå University of Technology, Course Coordinator and Instructor (6 months)
  Course: *Edge Computing in Robotics*
  Course: *6G in Automation*

2023
- **REBOOT SKILLS,** Digital Europe Programme, Course Coordinator and Instructor (1 year)
  Course: *Industrial Automation*

2021
- **AERO-TRAIN,** Marie-Sklodowska-Curie ITN - ETN project, Early-Stage Researcher (3 year)
  Project: *Augmented reality for enhancing semi-autonomous remote aerial manipulation*
- **5G EDGE INNOVATIONS FOR MINING,** Vinovva project, Ph.D. Researcher (3 year)
  Project: *Development of 5G / edge-based architectures for enabling autonomous drone operations in underground mines*
- **G-DRONES,** Vinovva project, Ph.D. Researcher (1.5 year)
  Project: *Integrate seamless communication between a base station (Grafana interface) and drones through MQTT protocol for safe aerial inspection of open-pit mines*

Research Publications

Journal Articles

1. V. N. Sankaranarayanan, **A. S. Seisa**, A. Saradagi, S. G. Satpute, and G. Nikolakopoulos, "Towards Safety and Scalability in the Coordination of Coupled Aerial-Ground Robots," *(submitted)*, 2026.
2. **A. S. Seisa**, S. Velhal, S. Kotpalliwar, S. Satpute, and G. Nikolakopoulos, "Optimization of Edge-Offloading for Centralized Controllers Through Dynamic Computational Resource Allocation," *IEEE Internet of Things Journal*, 2026. DOI: 10.1109/JIOT.2026.3650978.
3. A. Berra, J. Mellet, **A. S. Seisa**, *et al.*, "Towards Advanced Aerial Physical Inspection in Real Industrial Conditions," *(submitted)*, 2025.

4. M.-N. Stamatopoulos, P. Koustoumpardis, **A. Seisa**, and G. Nikolakopoulos, “Path Planning for Collaborative System of Tethered UAVs in Dynamic and Confined Environments,” *(submitted)*, 2025.
5. **A. S. Seisa**, B. Lindqvist, S. G. Satpute, and G. Nikolakopoulos, “An Edge Architecture for Enabling Autonomous Aerial Navigation with Embedded Collision Avoidance Through Remote Nonlinear Model Predictive Control,” *Journal of Parallel and Distributed Computing*, p. 104 849, 2024. DOI: `10.1016/j.jpdc.2024.104849`.
6. **A. S. Seisa**, B. Lindqvist, S. G. Satpute, and G. Nikolakopoulos, “E-CNMPC: Edge-Based Centralized Nonlinear Model Predictive Control for Multi-Agent Robotic Systems,” *IEEE Access*, vol. 10, pp. 121 590–121 601, 2022. DOI: `10.1109/ACCESS.2022.3223446`.
7. **A. S. Seisa**, S. G. Satpute, B. Lindqvist, and G. Nikolakopoulos, “An Edge-Based Architecture for Offloading Model Predictive Control for UAVs,” *Robotics*, vol. 11, no. 4, p. 80, 2022. DOI: `10.3390/robotics11040080`.

Conference Proceedings

1. G. Damigos, **A. S. Seisa**, N. Stathoulopoulos, S. Sandberg, and G. Nikolakopoulos, “Real-time Point Cloud Data Transmission via L4S for 5G-Edge-Assisted Robotics,” in *accepted*, 2026.
2. K. Nevala, S. B. Shahid, M. Tavangar, *et al.*, “Performance Improvements of 5G mmW Network with RIS in Underground Mine Environment,” in *2026 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit)*, IEEE, 2026, pp. 1004–1009. DOI: `10.1109/EuCNC/6GSummit68295.2026.11577357`.
3. **A. S. Seisa**, E. Pagliari, G. Damigos, E. Small, and G. Nikolakopoulos, “Real-World Deployment of a 5G-Connected Edge-Controlled Aerial Robot in Industrial Subterranean Mines,” in *2026 34th Mediterranean Conference on Control and Automation (MED)*, IEEE, 2026, pp. 521–526. DOI: `10.1109/MED70602.2026.11598005`.
4. **A. S. Seisa**, S. Kotpalliwar, S. G. Satpute, and G. Nikolakopoulos, “Dynamic Computational Resource Allocation for Ensuring Stability of Remote Edge-based Controlled Multi-agent Systems,” in *2025 European Control Conference (ECC)*, IEEE, 2025, pp. 2907–2913. DOI: `10.23919/ECC65951.2025.11187111`.
5. **A. S. Seisa**, V. N. Sankaranarayanan, G. Damigos, S. G. Satpute, and G. Nikolakopoulos, “Cloud-Assisted Remote Control for Aerial Robots: From Theory to Proof-of-Concept Implementation,” in *2025 IEEE 25th International Symposium on Cluster, Cloud and Internet Computing Workshops (CCGridW)*, 2025, pp. 171–176. DOI: `10.1109/CCGridW65158.2025.00032`.
6. A. Berra, V. N. Sankaranarayanan, **A. S. Seisa**, *et al.*, “Assisted Physical Interaction: Autonomous Aerial Robots with Neural Network Detection, Navigation, and Safety Layers,” in *2024 International Conference on Unmanned Aircraft Systems (ICUAS)*, IEEE, 2024, pp. 1354–1361. DOI: `10.1109/ICUAS60882.2024.10557050`.
7. J. Mellet, A. Berra, **A. S. Seisa**, *et al.*, “Design of a Flexible Robot Arm for Safe Aerial Physical Interaction,” in *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, IEEE, 2024, pp. 1048–1053. DOI: `10.1109/RoboSoft60065.2024.10522019`.
8. **A. S. Seisa**, S. G. Satpute, and G. Nikolakopoulos, “Cloud-Based Scheduling Mechanism for Scalable and Resource-Efficient Centralized Controllers,” in *IECON 2024 - 50th Annual Conference of the IEEE Industrial Electronics Society*, 2024, pp. 1–6. DOI: `10.1109/IECON55916.2024.10905254`.
9. G. Damigos, **A. S. Seisa**, S. G. Satpute, T. Lindgren, and G. Nikolakopoulos, “A Resilient Framework for 5G-Edge-Connected UAVs Based on Switching Edge-MPC and Onboard-PID Control,” in *2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)*, IEEE, 2023, pp. 1–8. DOI: `10.1109/ISIE51358.2023.10228114`.

10. V. N. Sankaranarayanan, G. Damigos, **A. S. Seisa**, S. G. Satpute, T. Lindgren, and G. Nikolakopoulos, "PACED-5G: Predictive Autonomous Control Using Edge for Drones over 5G," in *2023 International Conference on Unmanned Aircraft Systems (ICUAS)*, IEEE, 2023, pp. 1155–1161. DOI: `10.1109/ICUAS57906.2023.10156241`.

11. **A. S. Seisa**, N. Evangeliou, A. Tzes, and G. Nikolakopoulos, "Development and Experimental Evaluation of a 3DoF Tendon-Driven Probe for Robot Assisted Minimally Invasive Surgical Operations," in *2023 International Conference on Control, Automation and Diagnosis (ICCAD)*, IEEE, 2023, pp. 1–6. DOI: `10.1109/ICCAD57653.2023.10152334`.

12. M.-N. Stamatopoulos, P. Koustoumpardis, **A. Seisa**, and G. Nikolakopoulos, "Combined Aerial Cooperative Tethered Carrying and Path Planning for Quadrotors in Confined Environments," in *2023 31st Mediterranean Conference on Control and Automation (MED)*, 2023, pp. 364–369. DOI: `10.1109/MED59994.2023.10185884`.

13. **A. S. Seisa**, G. Damigos, S. G. Satpute, A. Koval, and G. Nikolakopoulos, "Edge Computing Architectures for Enabling the Realisation of the Next Generation Robotic Systems," in *2022 30th Mediterranean Conference on Control and Automation (MED)*, IEEE, 2022, pp. 487–493. DOI: `10.1109/MED54222.2022.9837289`.

14. **A. S. Seisa**, S. G. Satpute, B. Lindqvist, and G. Nikolakopoulos, "An Edge Architecture Oriented Model Predictive Control Scheme for an Autonomous UAV Mission," in *2022 IEEE 31st International Symposium on Industrial Electronics (ISIE)*, IEEE, 2022, pp. 1195–1201. DOI: `10.1109/ISIE51582.2022.9831701`.

15. **A. S. Seisa**, S. G. Satpute, and G. Nikolakopoulos, "A Kubernetes-Based Edge Architecture for Controlling the Trajectory of a Resource-Constrained Aerial Robot by Enabling Model Predictive Control," in *2022 26th International Conference on Circuits, Systems, Communications and Computers (CSCC)*, IEEE, 2022, pp. 290–295. DOI: `10.1109/CSCC55931.2022.00056`.

16. **A. S. Seisa**, S. G. Satpute, and G. Nikolakopoulos, "Comparison Between Docker and Kubernetes Based Edge Architectures for Enabling Remote Model Predictive Control for Aerial Robots," in *IECON 2022–48th Annual Conference of the IEEE Industrial Electronics Society*, IEEE, 2022, pp. 1–6. DOI: `10.1109/IECON49645.2022.9968933`.

Theses & Book Chapters

1. **A. S. Seisa**, "Towards the Next Generation Robotic Autonomy on the Edge: Scalability, Resiliency, and Dynamic Resource Allocation," in Luleå University of Technology, 2025.

2. **A. S. Seisa**, "Towards enabling the next generation of edge controlled robotic systems," in Luleå University of Technology, 2023.

3. **A. S. Seisa** and A. Koval, "Edge Connected ARWs," in *Aerial Robotic Workers*, Elsevier, 2023, pp. 245–253. DOI: `10.1016/B978-0-12-814909-6.00019-6`.

4. **A. S. Seisa**, "Development and control of an advanced robotic tool for enhanced minimally invasive robotic surgery," in University of Patras, 2020.

Academic & Professional Activities

Honors & Awards

2024 **IVA's 100 List** Reconfigurable 6G Quality of Service (QoS) and Edge Integration for Robotics and Automation

Academic & Professional Activities (continued)

Lecturing & Teaching

- 2025 — **AI in the Process Industry and Automation** (R0006E), Luleå University of Technology
- **Robotics for all course** (R0006E), Luleå University of Technology
- **Industrial Automation course** (R7008E), Luleå University of Technology
- 2024 — **Industrial Automation course** (R7008E), Luleå University of Technology
- **Robotics for all course** (R0006E), Luleå University of Technology
- 2023 — **Industrial Automation course** (R7008E), Luleå University of Technology
- **Robotics for all course** (R0006E), Luleå University of Technology
- 2022 — **Industrial Automation course** (R7008E), Luleå University of Technology

Talks

- 2025 — **Enabling 5G/Edge Connected Robots within Subterranean Environments**, Keynote at the IEEE International Symposium on Cluster, Cloud, and Internet Computing (CCGrid 2025) RISE Workshop, Trømso, Norway
- 2024 — **Cloud-enabled Remote Control**, Presentation at the AERO-TRAIN Summer School 2024, Human-Robot Interaction Day, Crete, Greece
- 2023 — **Edge Connected Drones**, Seminar at the Automatic Control Group 2023, Lund University, Lund, Sweden

Supervision

- 2025 — **Endemann, Moritz**, Multiple Edge Robots for World Mapping
- 2023 — **Markostamos, Georgios**, Autonomous Multi-Agent Exploration and Mapping
- 2022 — **Stamatopoulos, Marios-Nektarios**, Collaborative Control of Aerial Robots (drones) for Co-Manipulating Flexible Objects (ropes)
- **Dahlquist, Niklas**, Generation of Behavior Trees for Dynamic Agents Based on Market Inspired Task Allocation

Workshops & Events

- 2025 — **Full-Day CCGrid 2025 Workshop**, Committee at the "RISE" - Robotic Systems in the Edge-Cloud Continuum Workshop, Trømso, Norway
- 2024 — **Full-Day ICUAS 2024 Workshop**, Organizer at the Aerial Workers for Infrastructure and Asset Maintenance: The journey from "Lab" to "Real-World" Workshop, Crete, Greece
- **AERO-TRAIN Summer School 2024**, Presenter at the Aerial robots and their role in inspection and maintenance of industrial plants, Crete, Greece
- 2023 — **Mohamed Bin Zayed International Robotics Challenge 2023** Team ROC